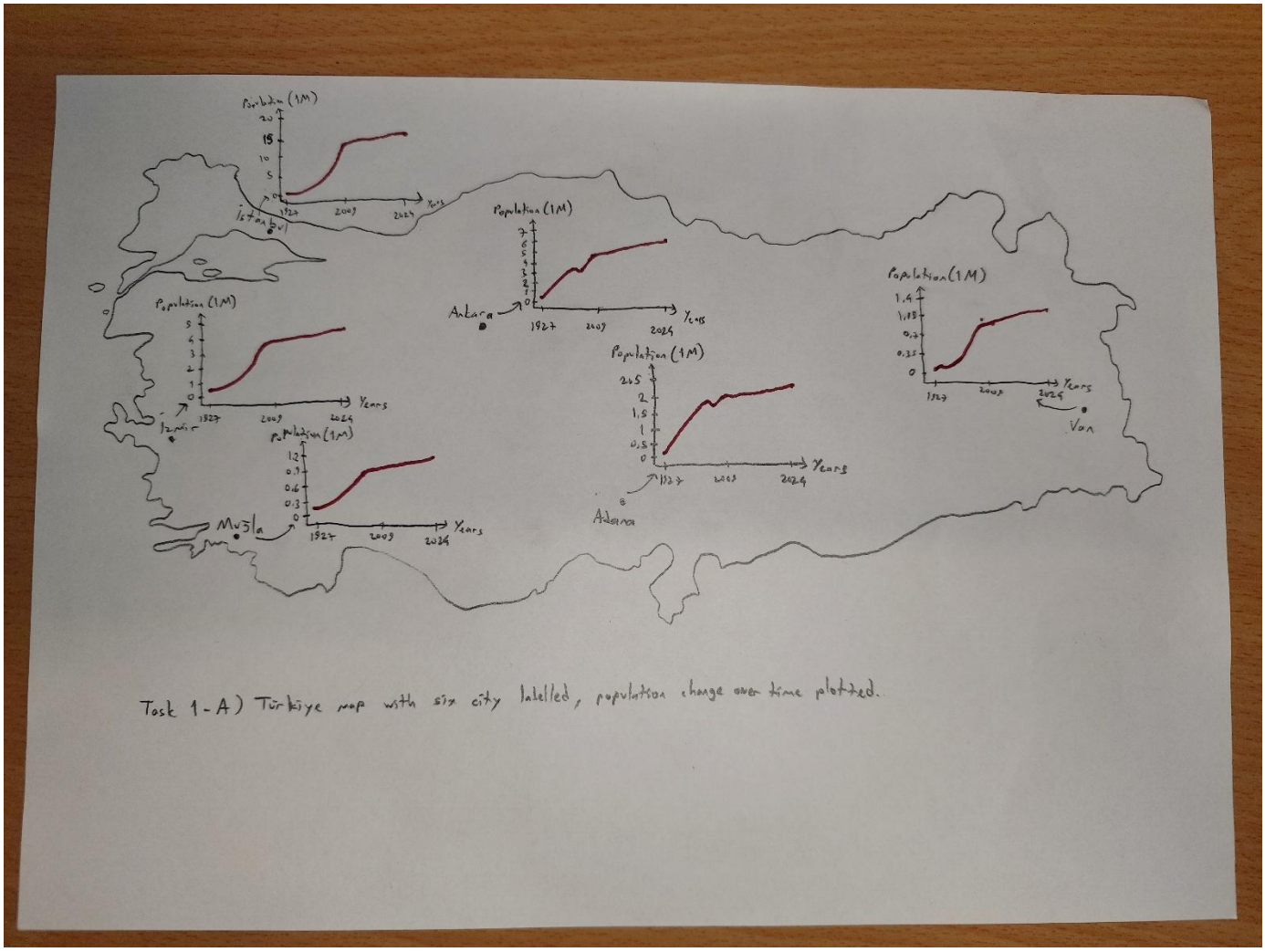
Yusuf Efdal Yılmaz – 33867 – DSA301 Project 1 – 16.03.2026

Github Link: https://github.com/beystef/dsa301-project1/

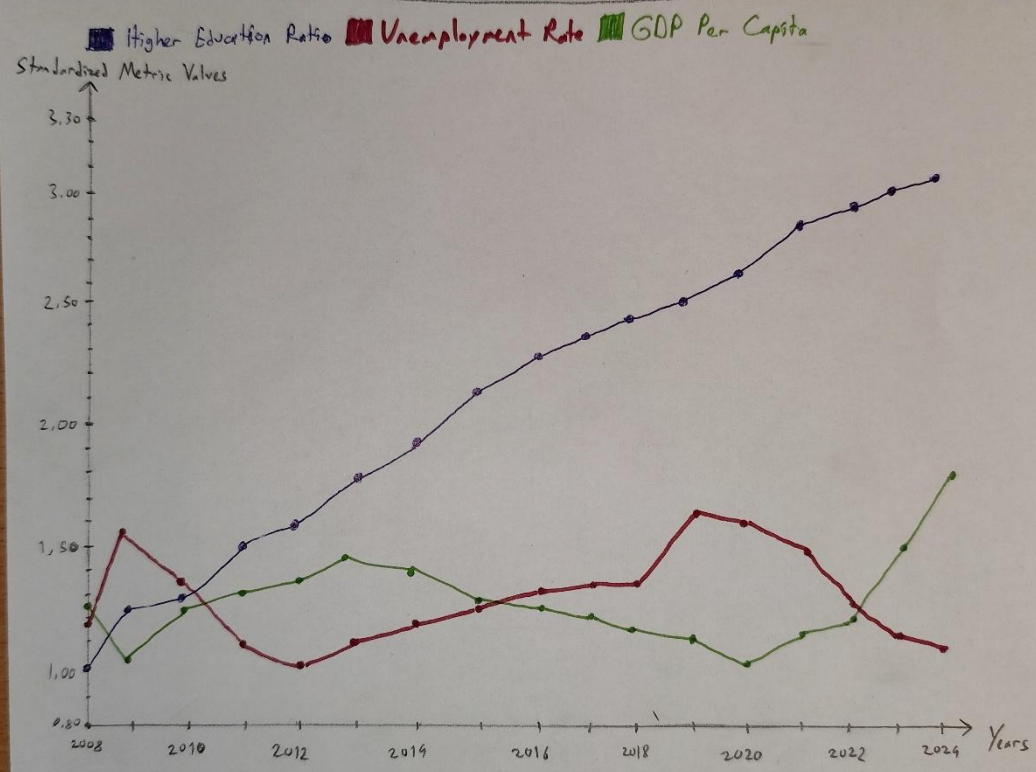


Task 1-A) Türkiye map with six city labelled, population change over time plotted.

Task 1-B) Three key socio-economic metrics plotted over time.

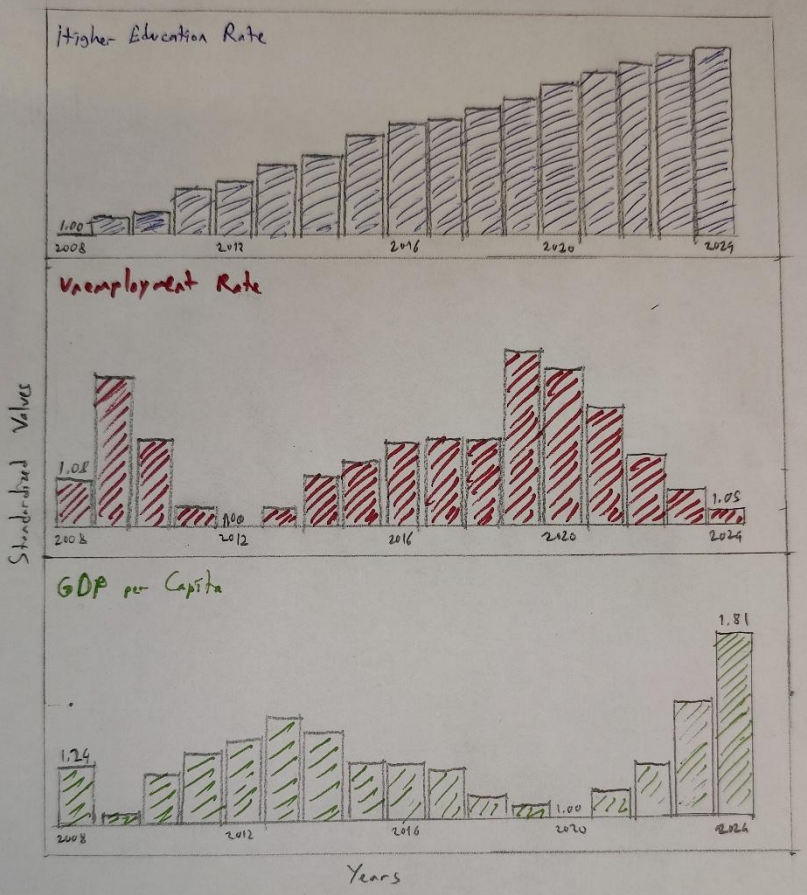
Question: How have GDP per capita, unemployment rate, and higher educated population ratio in Türkiye evolved over time, and what relationships can be visually observed between these indicators?
(Part C)

Key Metrics: GDP Per Capita, Unemployment Rate, Higher Educated Population Ratio.



Note: Each indicator values standardized by dividing by their minimum, making the smallest value = 1.
This approach let us easily understand the trend relation between metrics.

Task 2 | Sketch 1 : Small Multiples with Diverging Bar Columns



Rationale:

* I think, this sketch answers the question by showing the indicator changes in an obvious way by using bars, one can easily differenciate the heights.

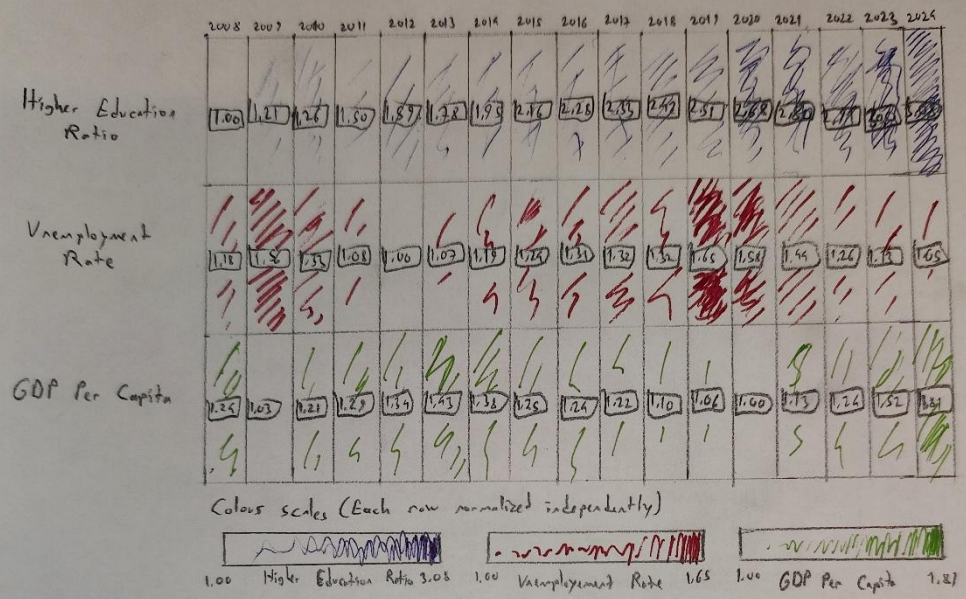
* The primary channel for data encoding is bar heights and coloring for different indicators.

* We can see that there is a "similarity" based on colors, also a "proximity" with panelled graphs. In addition, there is a "figure/ground" with the use of white background against colored bars.

* I think the foremost strength of the sketch is that the usage of heights as encoding. Viewers can easily make judgements on magnitudes.

* It has a weakness on cross comparisons than the multi-line chart.

Task 2 | Sketch 2: Heatmap



Rationale:

* This sketch answers the question by presenting indicator value change over time (left-to-right), and also enabling easy comparison between indicators in a vertial perspective.

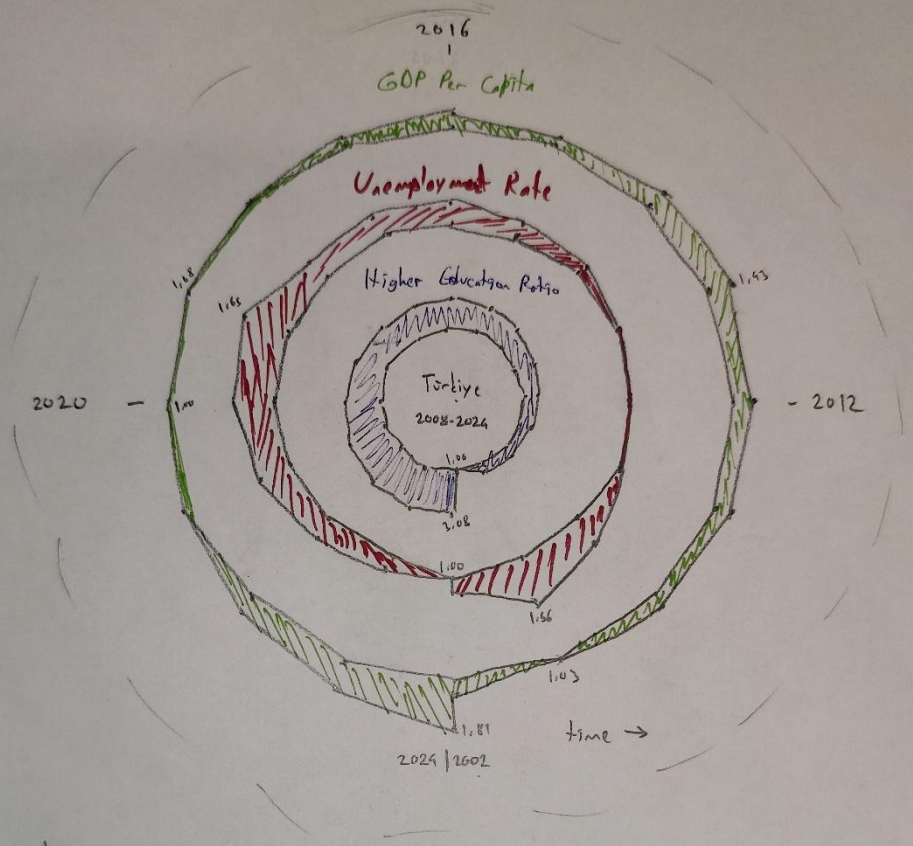
* Data encoding is different colors for each indicator with lighter being low value and darker being the high value.

* There is a "proximity" as each row grouped automatically aligned horizontally further more, there is "similarity" as overall darks are higher values and lighters are the lower values.

* The sketch is strong in terms of being very compact and easy to understand the patterns. However, it is weak on comparing the fluctuation margin between indicators since the rows are colored independently.

Task 2 | Sketch 3: Abstract Radial Clock

* Time flows clockwise from 6 pm ; 2024 → 2008



Rationale:

* I think abstract concepts are usually hard to grasp details when compared to the conventional sketchs such as line chart or bar chart. However, this sketch answers the question allowing viewer to understand the indicator value change by flowing time. One can compare the trend fluctuations looking at the standardized value waves.

* The sketch has the "continuity" as it has the circular sweep of each indicator. Furthermore, It has a nice "similarity" as it has three rings sharing same pattern. We can definitely say that it has a well "figure/ground" harmony.

* The data encoding is simple, different color hues for each of the indicator, radial filled area for the magnitude representation of the respective value.

* This sketch is strong being highly memorable and striking with the wavy flow while weaker on representing the quantitative aspects and radius comparison.